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كليـــــة الملك عبد الله الثاني للهندســـة

ELECTRIC MACHINES.

PROJECT

DESIGN OF A LIFTING MAGNET USING FEMM

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Abstract

This project focuses on designing a DC lifting magnet using the Finite Element Method Magnetics (FEMM) software. The primary objective is to develop a lightweight and efficient lifting magnet that meets specified design requirements while minimizing losses. Through this assignment, students will gain practical experience with finite-element techniques and FEMM, exploring both its advantages and limitations.
The design process involves creating an initial concept based on fundamental electromagnetics, followed by iterative optimization in FEMM to achieve the desired performance. Key deliverables include the magnet's estimated weight, losses, and the required coil voltage to hold a given mass at specified air gaps (1mm and 5mm). Additionally, the variation of coil inductance with air gap will be analyzed to inform system voltage requirements during operation. This hands-on exercise emphasizes creativity and individual problem-solving, reinforcing the understanding of electromagnetic design principles while preparing students for practical engineering challenges.

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INTRODUCTION

This project focuses on utilizing FEMM (Finite Element Method Magnetics) software to design a simple DC lifting magnet capable of lifting a 200kg weight. The primary goal is to familiarize myself with the FEMM software and the principles of finite element analysis in the context of designing electromagnetic devices. Through this process, I aim to demonstrate the power and advantages of using electromagnetic finite element analysis software for developing efficient and effective designs. The project will involve creating a high-powered DC lifting magnet, leveraging my understanding of electromagnetics to ensure the magnet can safely and effectively lift a 200kg load.

OBJECTIVES

Magnet Specification

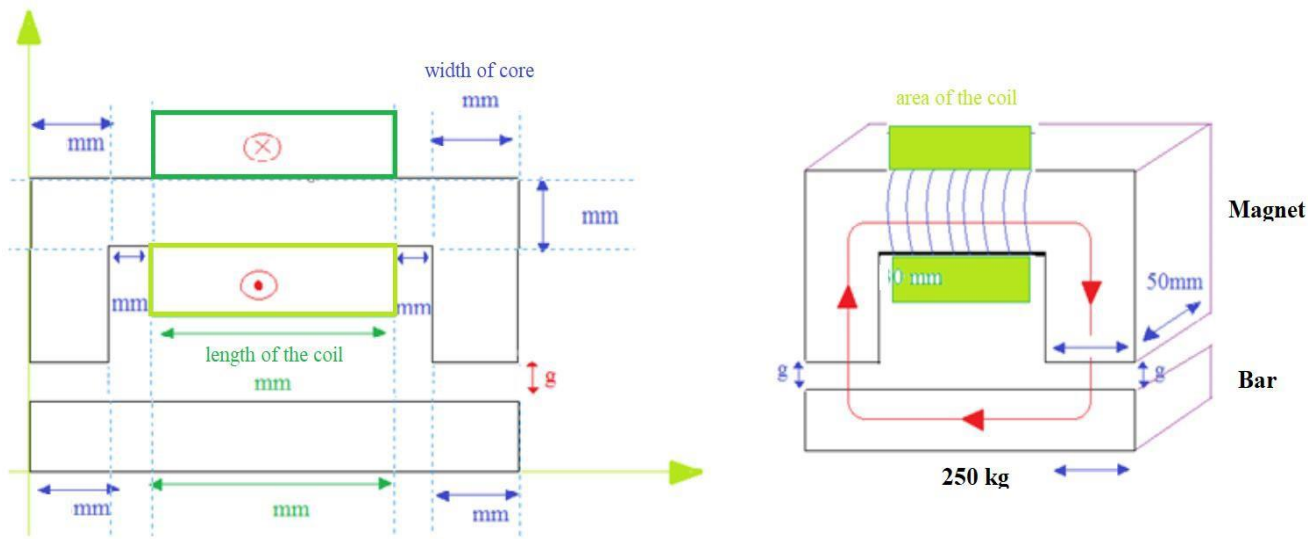
The magnet is designed to lift a mass of 200 kg using a steel lifting bar (rectangular) from a distance of 5mm and then hold it against a non-magnetic spacer of 1mm thick. You need to design the magnet to lift the 200 kg mass and hold it against the spacer subject to the constraints below.

- DC – Single winding magnet
- Copper coil
- Magnetic steel – M19 (from FEMM materials library)

Design Constraints

- Max coil current density at 5mm gap – J = 7A/mm2 (overload)
- Max coil current density at 1mm gap – J = 5A/mm2 (steady-state)
- Coil copper fill factor – 75%
- The maximum current in the coil <5 A.

PROCEDURE AND METHODS



Suggested Design of the Lifting Magnet

- **Step 1:** Start by the value of the required force (F_{req}) that would allow the magnet to be able to lift this weight, this is usually the weight of the bar that we need to lift, and you may add a 10% extra weight to guarantee the design, using the following formula:

$$F_{required} = Weight \pm (10\%) \quad (1)$$

Weight=m*g

F_{req}=(Weight+10%)=(200*9.8+10%)=2156 N.

- **Step 2:** Start by correlating the area of the coil (windings) using the following formula:

$$A_{coil} = \frac{\left(\frac{\overline{N \times I}}{J}\right)}{Fill\ Factor} \qquad (2)$$

- N: Number of turns
- I: current in Ampere
- J: Current density A/mm^2

Note: I picked J=7A/mm2

Pick value for the current this value should be < 5A.

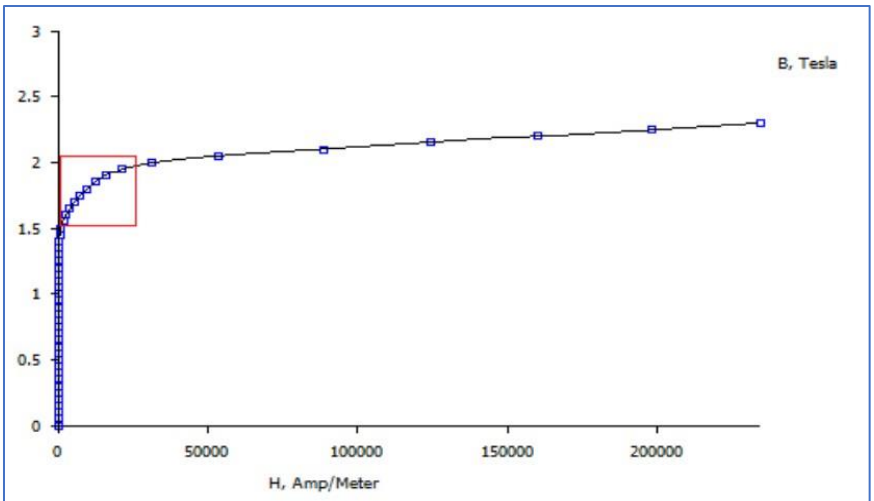
So, I choose the value **4.9A** to reduce the number of turns of the coil, then:

$$A_{Coil} = \frac{\frac{N \times I}{J}}{Fill\ Factor} = (N \times 0.9333\ )$$

- **Step 3:** Draw the B-H curve for the material of the core (e.g. the figure below shows the B-H curve for M-19 Steel). Using the figure, pick an appropriate value for the required flux density B.

Restrictions: the selected B should meet the following:

- Avoid saturation region.
- To allow to make the most of the steel material.
- To reduce the area of the core steel
- (To reduce the cost).



B-H curve for the M-19 steel

By Picking the value for the flux density from the graph of the M19 material, the value should be between 1.5 and 2 Tesla,
<u>Picking 1.5T</u>.

- **Step 4:** Estimating the area of the air gap using the following formula (the force attraction formula):

$$B_{required} = \sqrt{\frac{2\mu_0 F_{required}}{A_{air\,gap}}} \qquad (3)$$

$$A_{airgap} = \frac{2\mu_o \times F_{Req}}{B^2}$$

$$A_{airgap} = \frac{2\pi \times 4 \times 10^{-7} \times 2156}{1.5^2}$$

$$A_{airgap} = 2.408 \times 10^{-3}\ m^2$$

$A_{airgap} = 2408\, mm^2$

- **Step 5:** Choosing the required number of turns N, using the following formula (Ampere's law):

$$l.H = NI \qquad (4)$$

Ignoring core material by assuming $\mu_r = \infty$
hence:

$$2lg.\left(\frac{B_{required}}{\mu_0}\right) = NI \quad (5)$$

(lg) is the length of the air gap which is **5mm.**

$$N = \frac{2L_g\left(\frac{B_{req}}{\mu_o}\right)}{I} = \frac{2\times5\times10^{-3}\left(\frac{1.5}{4\pi\times10^{-7}}\right)}{4.9} = 2436 \text{ turn}$$

So, (5) can gives estimated number of turns.
Hint: Consider the worst-case scenario (Maximized current within the range)

- **Step 6:** Re-calling (2), and substitute the value of the turns, we get the coil area in mm^2.

  N=2984 turn

$$A_{Coil} = \frac{\frac{N \times I}{J}}{Fill\ Factor} = (N \times 0.9333\ )\ mm^2$$

$$A_{Coil} = (2436 \times 0.9333\ )\ mm^2$$

$$A_{Coil} = 2274\ mm^2$$

We have obtained the following so far:

$A_{Coil} = 2274\ mm^2$

$A_{airgap} = 2408\ mm^2$

$L_g = 5\ mm$

N =2436 turn

Depth = 50 mm

- **Step 7:** Use the A_{airgap} and the depth of the core (50 mm), to calculate the width of the air gap (note this should be same with the width of the core).

$$W_g = W_{core} = \frac{A_{airgap}}{Depth} = \frac{2408}{50} = 48.16\ mm$$

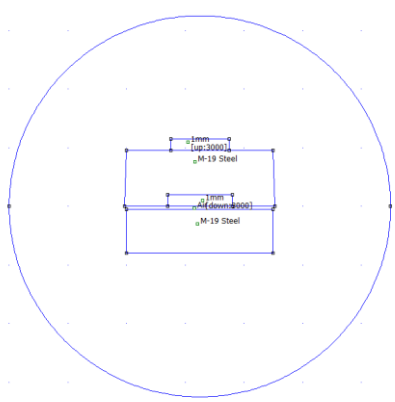
- **Step 8:** Pick length and width of the coil in such a way to get the previously calculated area.

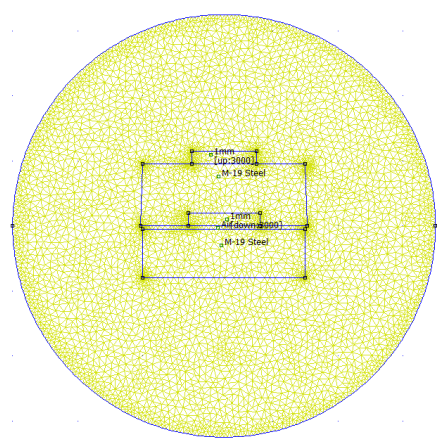
  Assume $W_{coil} = 50\ mm$ to match the dimensions:

$$L_{coil} = \frac{A_{coil}}{W_{coil}} = 45.5\ mm$$

- **Step 9+10:** Simulation and Checking the design.

TRAIL (4)

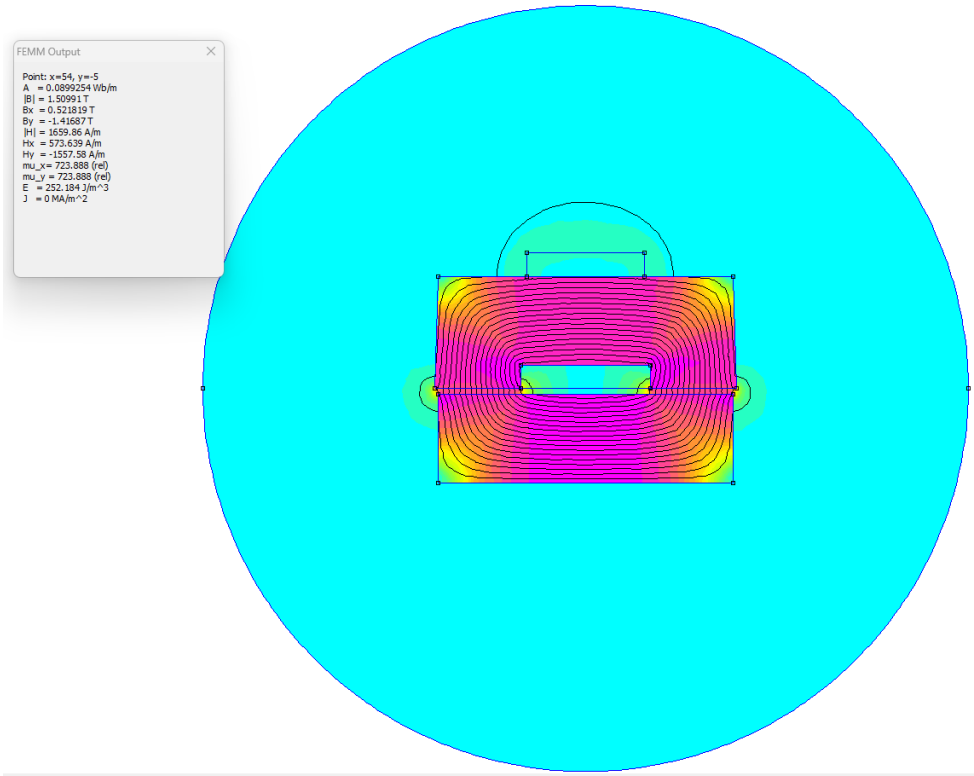


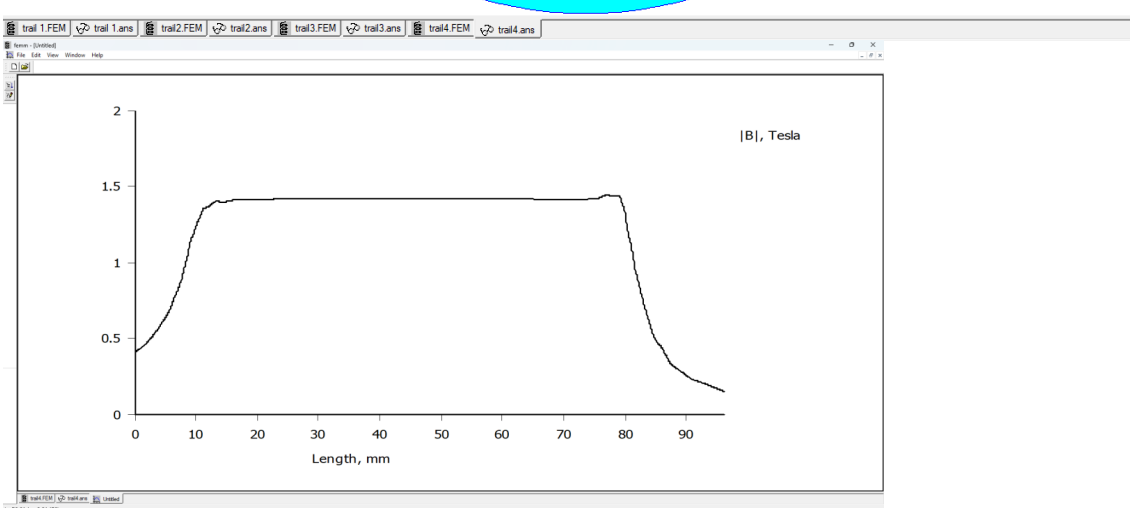


I needed to Increase the #turns so I assumed that N=3000:

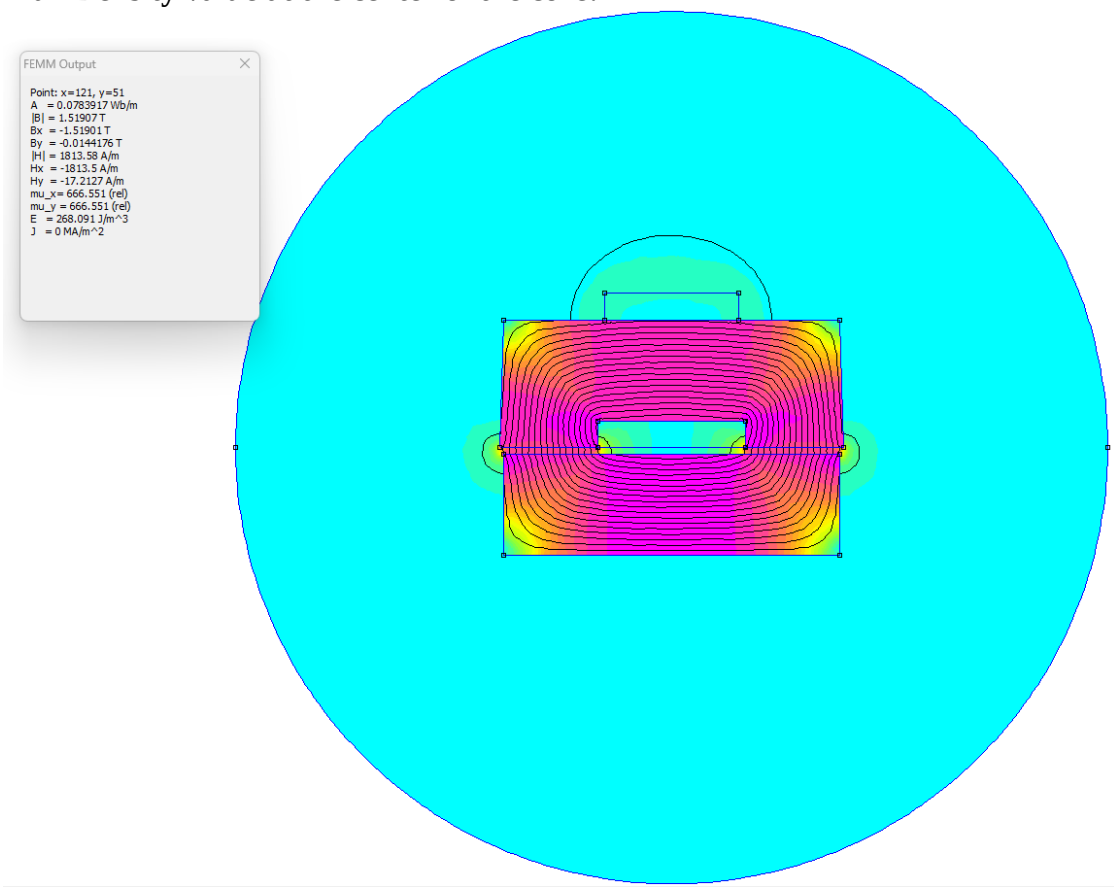
$L_g = 5\ mm$ N =3000 turn Depth = 50 mm I=4.9 A

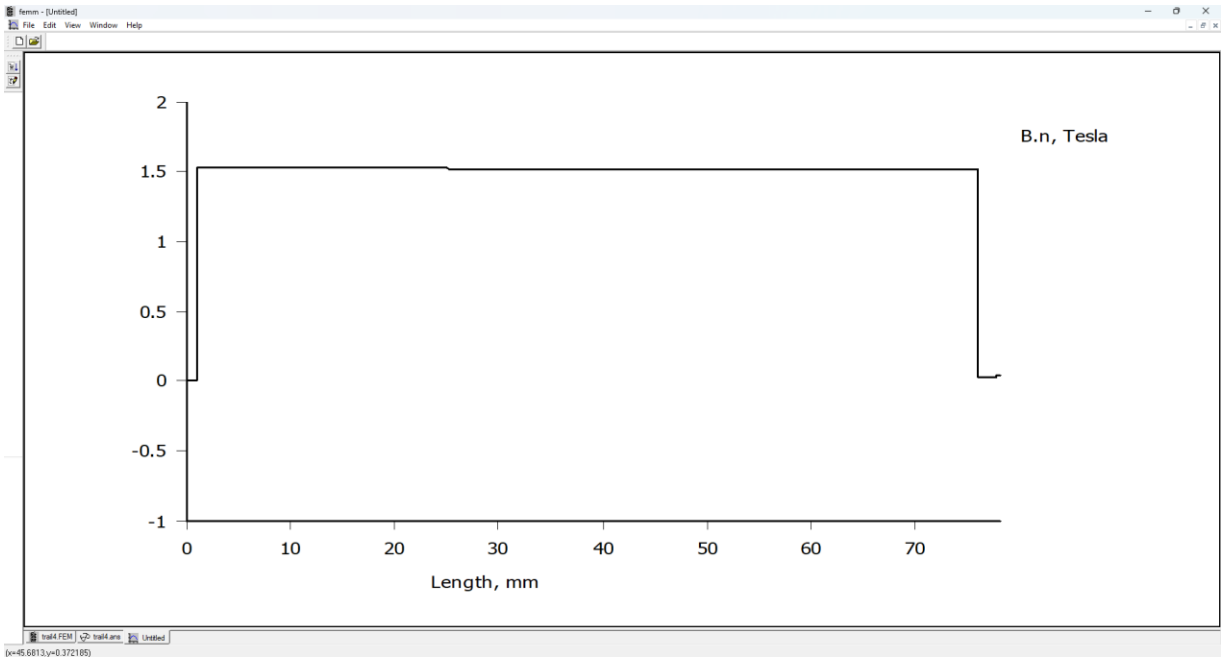
Flux Density value at the Air gap:



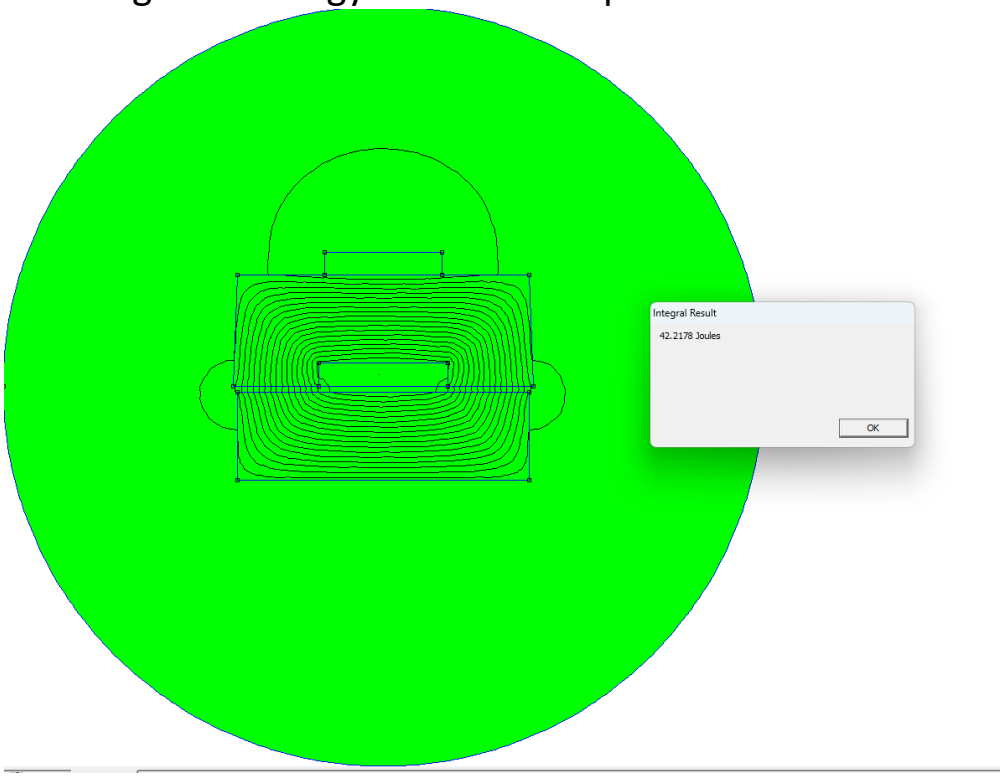


Flux Density value at the center of the core:



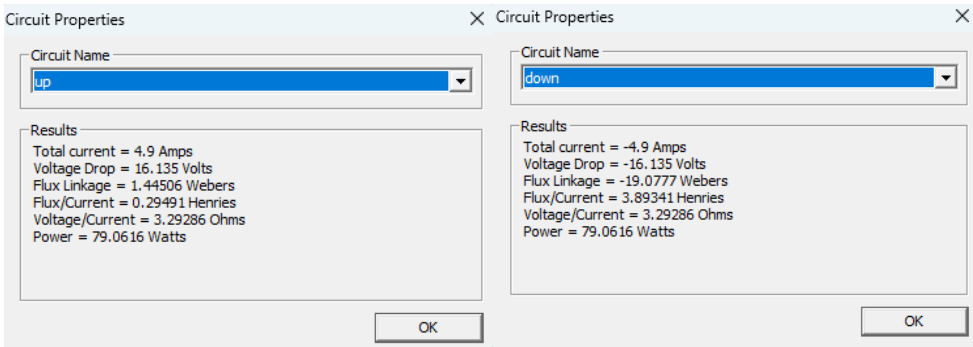


The magnetic energy stored in the problem:



Energy stored =42.2178 Joules

The self-inductance of the coil:



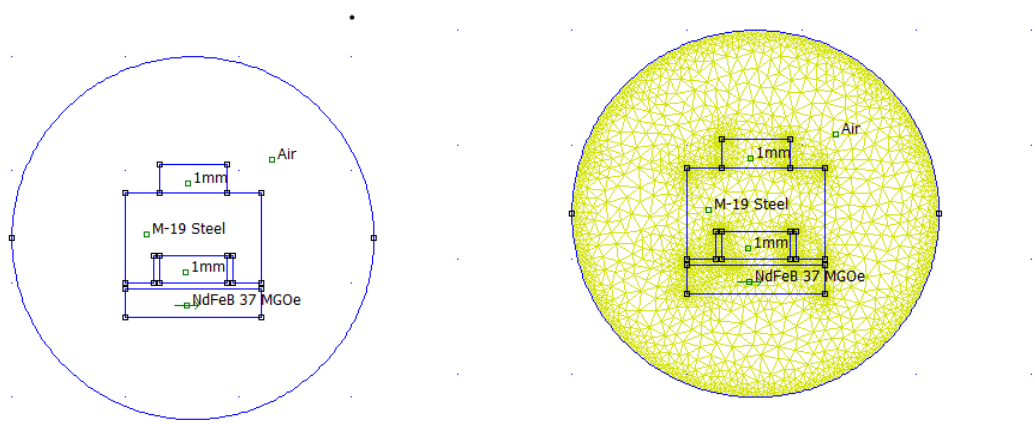
The self-inductance = Up + Down= 0.295+3.893=4.188 Henries

RESULTS AND DISCUSSIONS

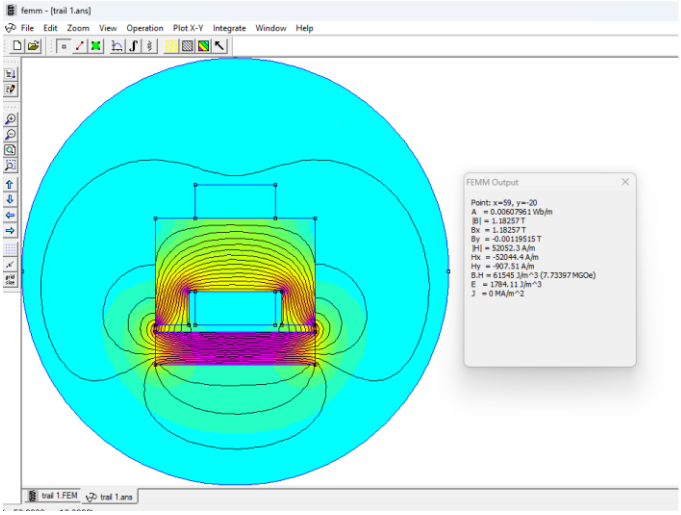
Trail (1)

Assuming 80x60 dimension and $L_g = 5\ mm$ N =2436 turn Depth = 50 mm I=4.9 A

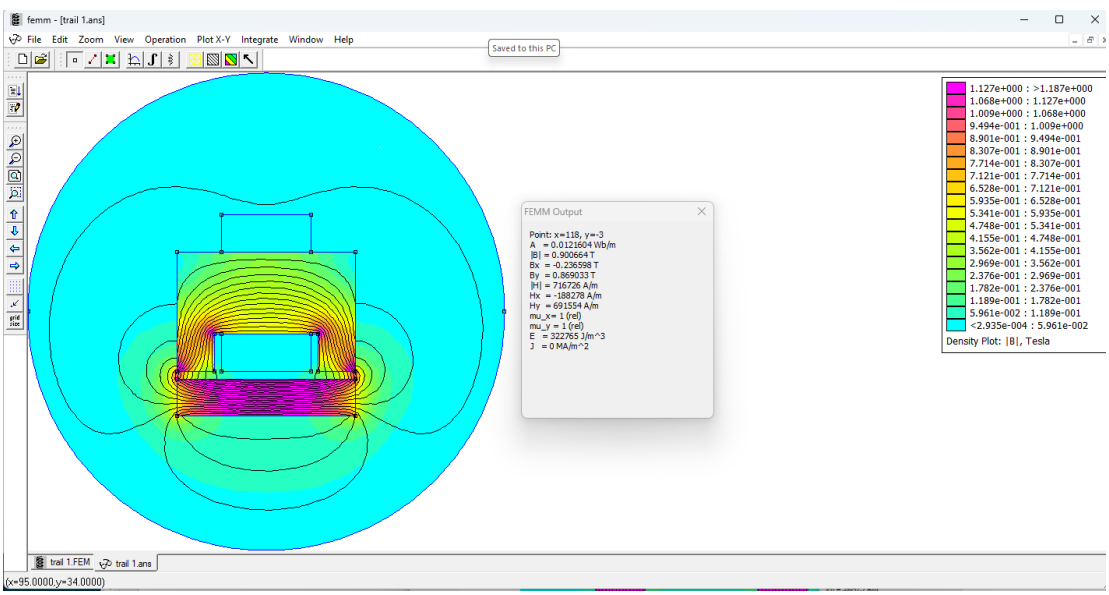
I focused on the flux density at first and after that I can deal with the turns



Flux Density value at the center of the core:



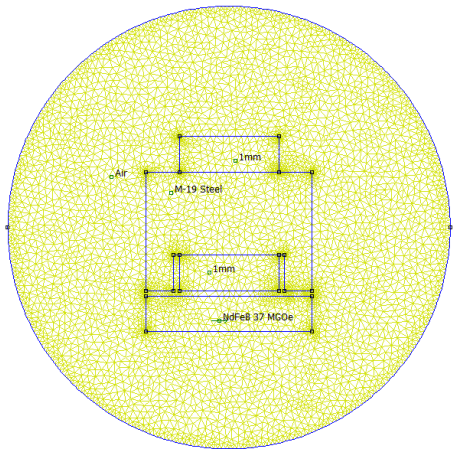
Flux Density value at the Air gap:



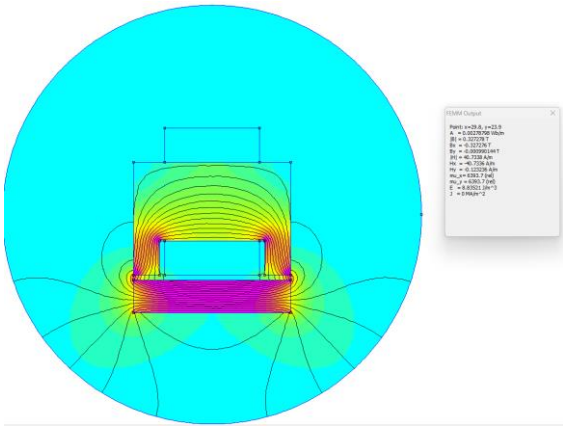
Trail (2)

Changes the length and the width of the circuit to hopefully have a different flux density:

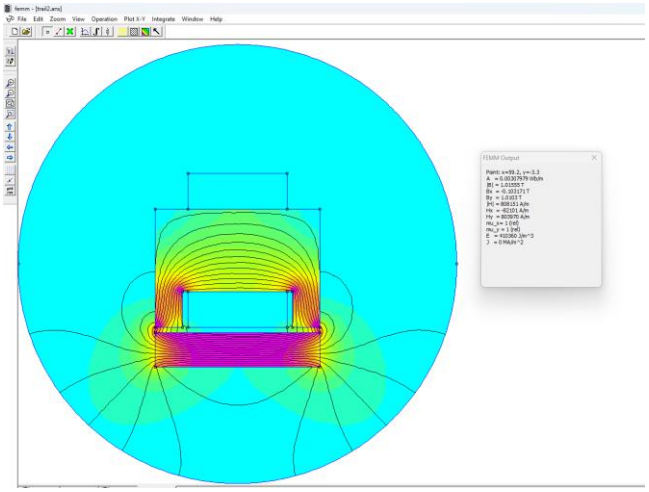
$L_g = 5\ mm$ N =2436 turn Depth = 50 mm I=4.9 A



Flux Density value at the center of the core:



Flux Density value at the Air gap:

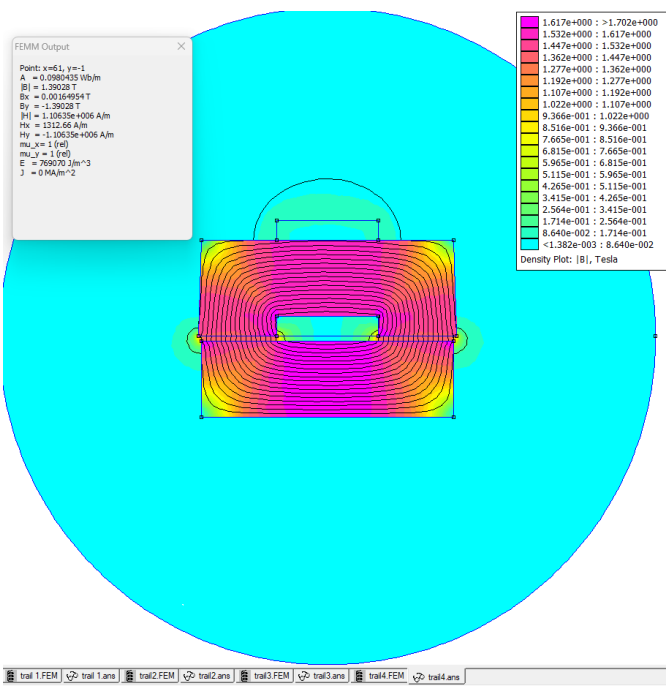


Still need to change the shape of it

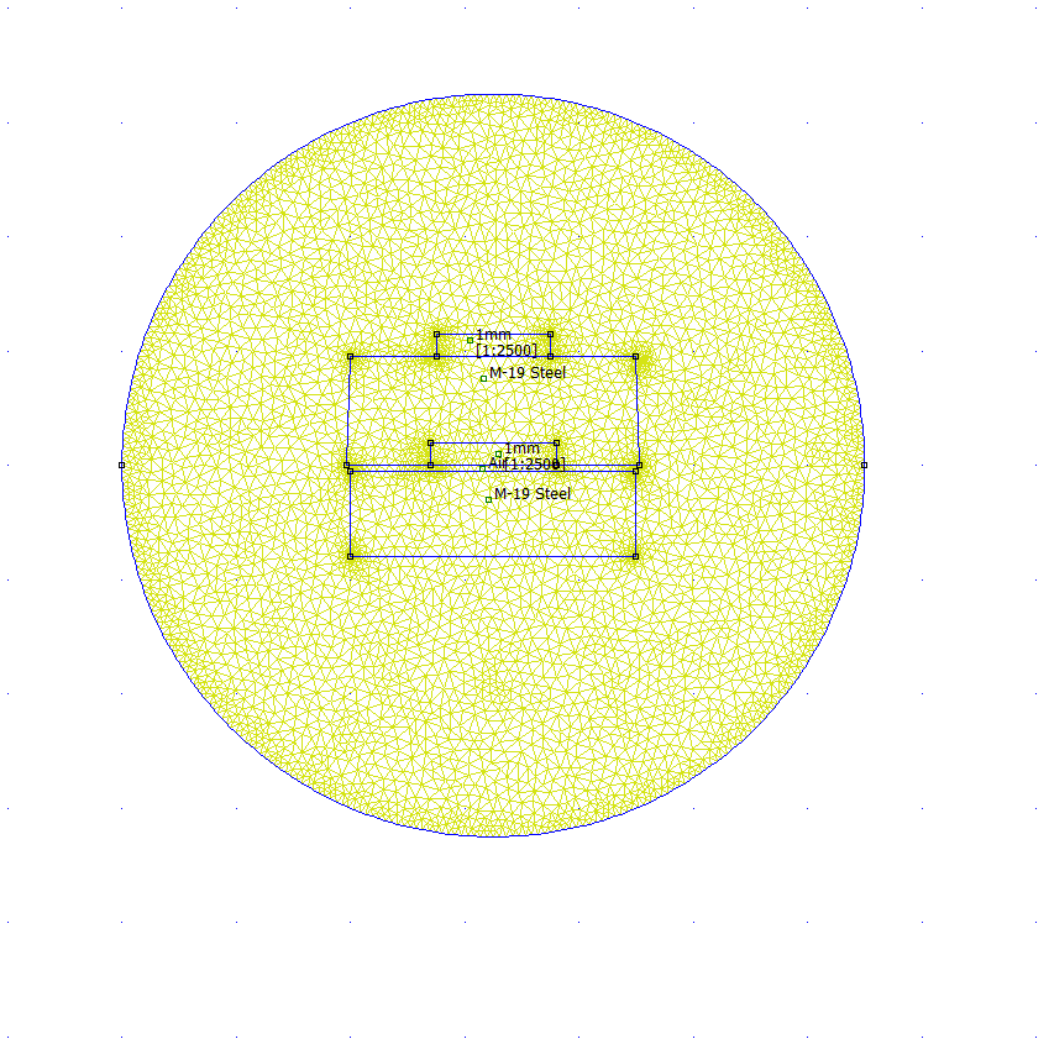
Trail (3)

Trying to change the entire design, after many tries it began to be close to B=1.5 T.

$L_g = 5\ mm$ N =2436 turn Depth = 50 mm I=4.9 A



Trail (4)



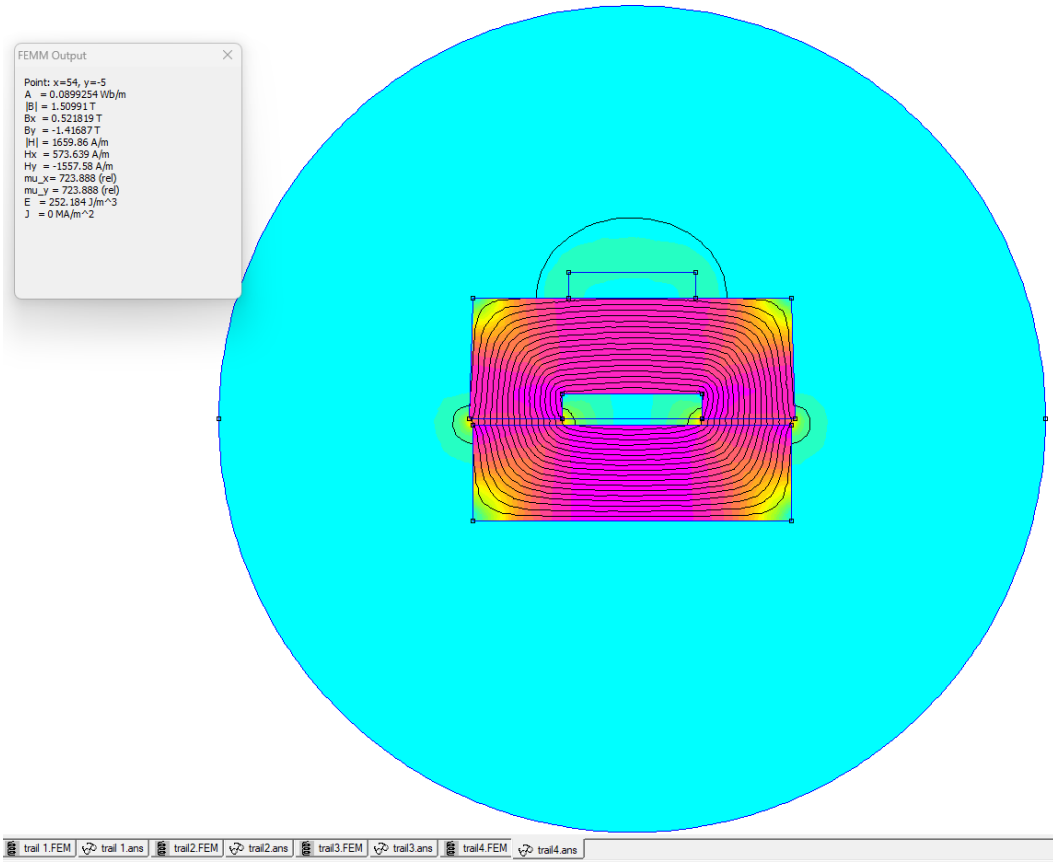
I added a little curve in the width with some dimensions change. hopefully I get the flux lines I want and it worked!

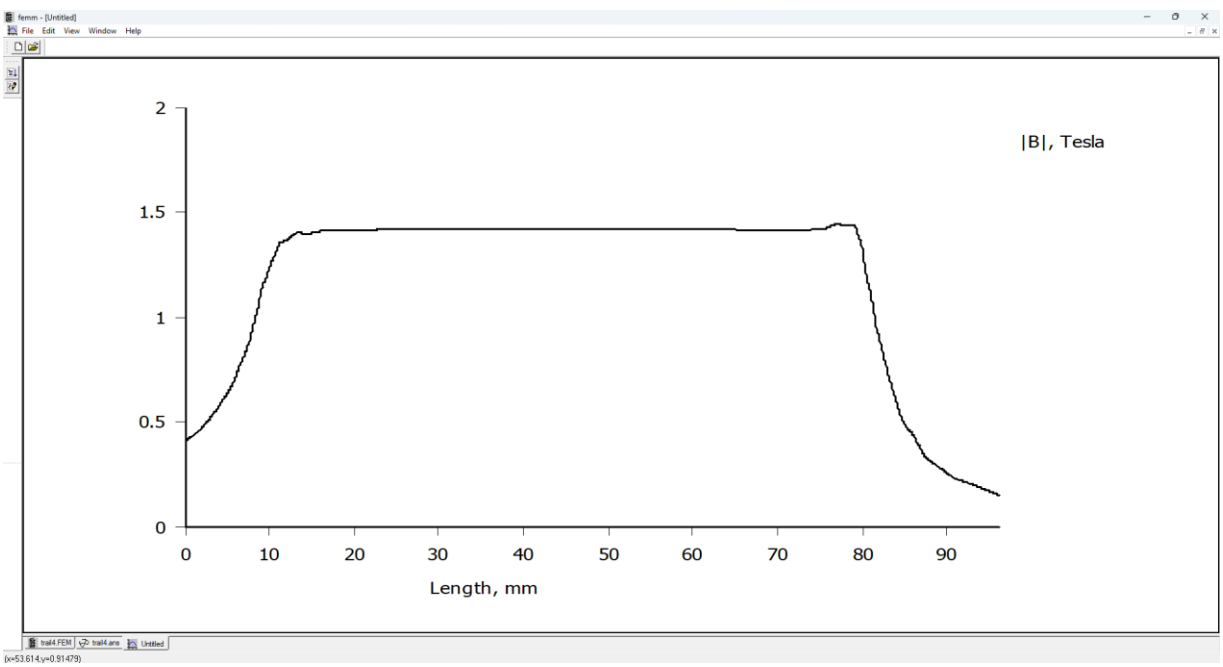
I also change the N to be 3000 to get the best results.

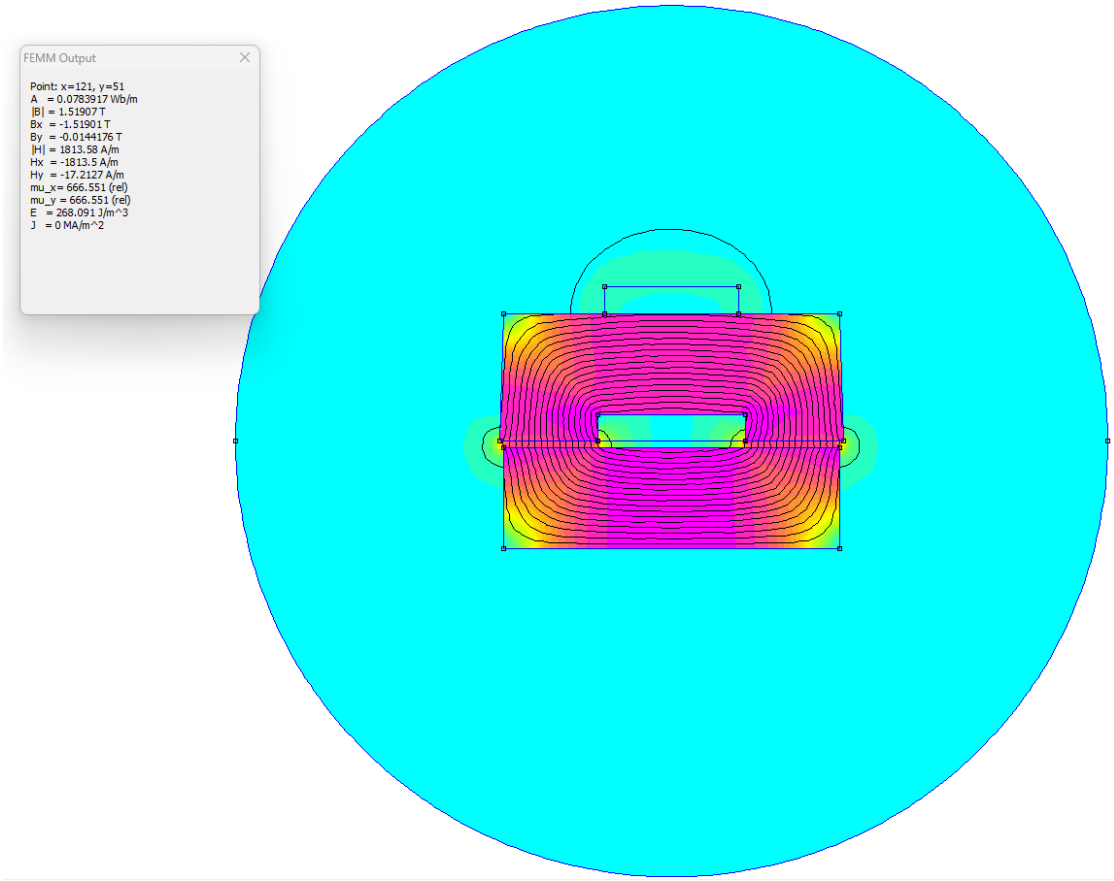
$L_g = 5\ mm$ N =3000 turn Depth = 50 mm I=4.9 A

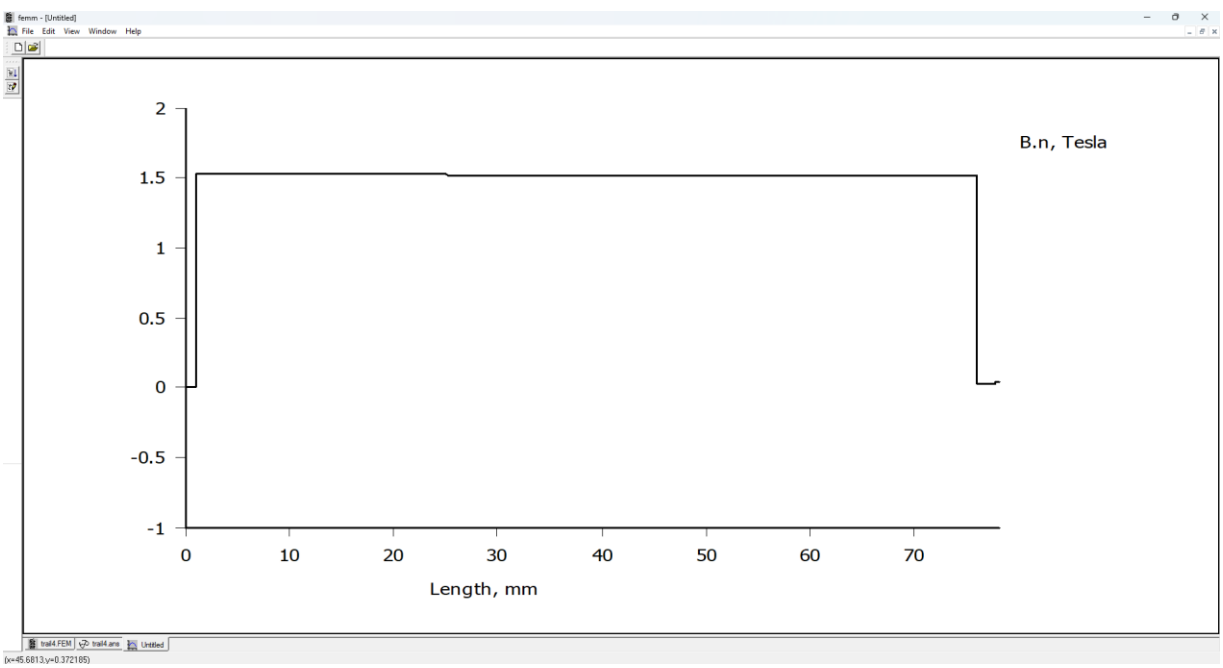
Using 100x250 mm for the magnet

I get the result B=1.5 T


FEMM Output
Point: x=54, y=-5
A = 0.0899254 Wb/m
|B| = 1.50991 T
Bx = 0.521819 T
By = -1.41687 T
|H| = 1659.86 A/m
Hx = 573.639 A/m
Hy = -1557.58 A/m
mu_x= 723.888 (rel)
mu_y = 723.888 (rel)
E = 252.184 J/m^3
J = 0 MA/m^2
trail 1.FEM
trail 1.ans
trail2.FEM
trail2.ans
trail3.FEM
trail3.ans
trail4.FEM
trail4.ans


femm - [Untitled]
File Edit View Window Help
|B|, Tesla
2
1.5
1
0.5
0
0
10
20
30
40
50
60
70
80
90
Length, mm
trail4.FEM
trail4.ans
Untitled
(x=53.614,y=0.91479)


FEMM Output
Point: x=121, y=51
A = 0.0783917 Wb/m
|B| = 1.51907 T
Bx = -1.51901 T
By = -0.0144176 T
|H| = 1813.58 A/m
Hx = -1813.5 A/m
Hy = -17.2127 A/m
mu_x= 666.551 (rel)
mu_y = 666.551 (rel)
E = 268.091 J/m^3
J = 0 MA/m^2


femm - [Untitled]
File Edit View Window Help
B.n, Tesla
2
1.5
1
0.5
0
-0.5
-1
0
10
20
30
40
50
60
70
Length, mm
trail4.FEM
trail4.ans
Untitled
(x=45.6813,y=0.372185)

CONCLUSIONS

In conclusion, Using FEMM helps with accurate solving for the magnetic circuits we design ,it is also good for choosing materials to calculate the overall cost , designing a project in FEMM helps to understand the relationships between the dimension of the circuit , the number of coils and the value of current, simulation in FEMM helps to understand all variables and check the design we do before making any steps in real life ,to ensure that our work is sufficient.

REFERENCES

https://www.femm.info/wiki/HomePage